01 — AML PROJECT · PHASE 2

SEQUENCE AND HYBRID MODELS FOR ENGINE LIFE

Do deep models that read the raw sensor history beat engineered features at predicting remaining useful life? Tested on NASA's C-MAPSS FD001, with FD002 and FD004 for operating regimes.

TEAM

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Phase 1 left one question open

27.55

Phase 1 test RMSE: engineered Random Forest, uncapped target

16.21

The same model with training RUL capped at 125: the cap mattered more than the model

The gap: our features summarise only the last 5, 10 and 20 cycles. A model that reads the raw sequence might find longer patterns.

Sequence models

LSTM, temporal CNN and Transformer on 30-cycle windows of raw sensors.

Hybrids

Do they beat their best single branch? A value test fixed in advance.

Ablations

Does longer history help, and at which stage of an engine's life?

FD002 and FD004

Six operating regimes: does scaling within each regime help?

Published FD001 scores use different rules

Paper, model	RMSE	Cap	Window	Runs
Zhang 2022, DAST	11.43	125	40	10
Jin 2022, Bi-LSTM	12.58	125	30	not stated
Galchar 2026, LSTM	14.93	130	30	1 seed
Chen 2021, LSTM	15.43	n.s.	30	10
Gupta 2026, LSTM	15.97	125	30	5 seeds, ± 0.45
Ours, Phase 1 RF	16.21	125	—	1 seed

Every value read from the paper's own results table. n.s. = not stated. Galchar and Gupta are preprints.

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don't say whether test labels were capped too, which changes RMSE

1

reports a spread over seeds. So we compare models only inside our own protocol

The cap is chosen from training engines

1 · Find where wear starts

A health index per engine, and a hinge fit: flat, then rising toward failure. The RUL at the bend is the onset.

2 · Score where it matters

K = 90th percentile of onset RULs (folds: 134, 132, 123), so **K = 130**. Selection RMSE uses held-out cycles with true RUL ≤ K.

RANDOM FOREST, SELECTION RMSE BY TRAINING CAP

Cap	RMSE ≤ K, 5 seeds
100	15.62 ± 0.36
125, kept	14.97 ± 0.30
150	17.42 ± 0.36
none	23.57 ± 0.70

Every ML model was then re-tuned on this metric. The bar for deep learning: **XGBoost on engineered features, 14.49** in cross-validation.

30-cycle windows; regularization changed nothing

One window per cycle

The last 30 cycles of 15 sensors, scaled on training engines only. Short histories are padded with the engine's first cycle, so every cycle gets a prediction.

`20,631 windows · 17,731 without padding`

Split by engine

Phase 1's 3 engine folds, plus held-out engines inside each fold for early stopping. No engine is on both sides.

TUNED LSTM, ONE CHANGE AT A TIME · SELECTION RMSE, 3 FOLDS

Setting	RMSE ≤ K
Baseline (dropout 0.3)	15.78 ± 0.44
Weight decay 0.0001	15.83 ± 0.39
Input noise 0.01	15.87 ± 0.40
Dropout 0.1	16.81 ± 0.42
Training stride 2	17.47 ± 1.03
Weight decay 0.001	17.60 ± 1.00

11 settings in all; none beat the baseline, so the tuned LSTM stays as it is.

Three encoders, one contract

LSTM

Reads the window in order and carries a memory forward. The standard baseline for degradation over time.

`1 layer · 64 units`

`CV 15.78 ± 0.44`

Temporal CNN

Causal convolutions with growing gaps: sees 61 cycles back in parallel, no recurrence.

`32 channels · 4 levels`

`CV 17.24 ± 1.19`

Transformer

Self-attention lets any cycle in the window draw on any other, then averages them.

`1 layer · 4 heads · d 32`

`CV 15.59 ± 0.23`

INPUT	ENCODER	HEAD	SAME FOR ALL
[batch, 30 cycles, 15 sensors]	→ one embedding per window	→ one RUL value	Trainer, early stopping, CPU, seed 42

Two hybrids, each against its best branch

A · Gated fusion

An LSTM branch and an engineered-feature branch, mixed by a learned gate: **z = g · sequence + (1 − g) · features**.

`CV 16.59 vs LSTM 16.87 · 3 seeds`

Passes criterion 3 only: seed spread 0.24 against 0.98.

B · Two-stage

XGBoost on the engineered features plus LSTM embeddings, cross-fitted so no engine's embedding comes from a model that saw it.

`CV 14.65 vs XGBoost 14.67 · 3 seeds`

Null result: the embeddings add nothing measurable.

VALUE TEST, FIXED IN ADVANCE	1 · BOOTSTRAP	2 · WILCOXON	3 · STABILITY
A hybrid adds value only if one holds:	95% CI of the difference below 0	Holm-corrected $p < 0.05$	Lower seed spread, mean no worse

Longer history does not help

ONE CHANGE AT A TIME · RMSE CHANGE VS BASE, PAIRED OVER ENGINES

Variant	Seeds	Change	Verdict
Window 15 / 60 / 90	3	+0.02 to +0.07	none
Feature branch only	3	+0.79	borderline
Sequence branch only	3	+0.29	none
No cap	1	+14.32	worse
Cap 150	1	+3.15	worse
Cap 100	1	−1.13	none
Asymmetric loss	1	−0.14	none

Real = 95% CI excludes 0 and Holm-corrected $p < 0.05$ over 14 comparisons. Feature branch only: CI excludes 0, $p = 0.051$.

0.07

Largest change from any window length, at every life stage: smaller than the seed spread

−12%

NASA score with the asymmetric loss: it shrinks late errors, RMSE unchanged

The official test cannot separate the models

PAIRED OVER 100 TEST ENGINES · PAIRINGS FIXED BY CROSS-VALIDATION

A vs B	A − B, 95% CI	Verdict
Hybrid A vs LSTM	−0.54 [−3.35, +2.24]	null
Hybrid B vs XGBoost	−0.53 [−1.45, +0.38]	null
Transformer vs XGBoost	−0.11 [−2.42, +2.26]	tie

Every Holm-corrected p = 1. Each model's own CI is 4–5 cycles wide; the top seven are within 1.1. Hybrid A's cross-validation pass on stability does not carry over: on the test it varies more than the LSTM (0.46 vs 0.33).

FD002 / FD004 · SIX REGIMES

−3.20

LSTM RMSE change from per-regime scaling on FD004 in cross-validation (FD002: −1.30)

22–27%

of test engines are above the cap. They carry most of the test error (44–50 vs 15–17 RMSE)

Every top model lands near 15.5 RMSE

Model, cap 125	RMSE	NASA	Late RMSE (≤ 50)
Hybrid B, two-stage	15.18 ± 0.55	356	6.16
Hybrid A, gated	15.40 ± 0.46	488	5.08
Transformer	15.60 ± 0.35	443	6.44
XGBoost, engineered	15.71	389	6.27
Temporal CNN	15.83 ± 0.55	414	7.58
LSTM	15.95 ± 0.33	499	5.50
Random Forest, engineered	16.21	412	7.39

One run at the model freeze. Networks: mean ± s.d. over 5 seeds; ML: 1 seed. Scored once per engine, at its last cycle.

5–8

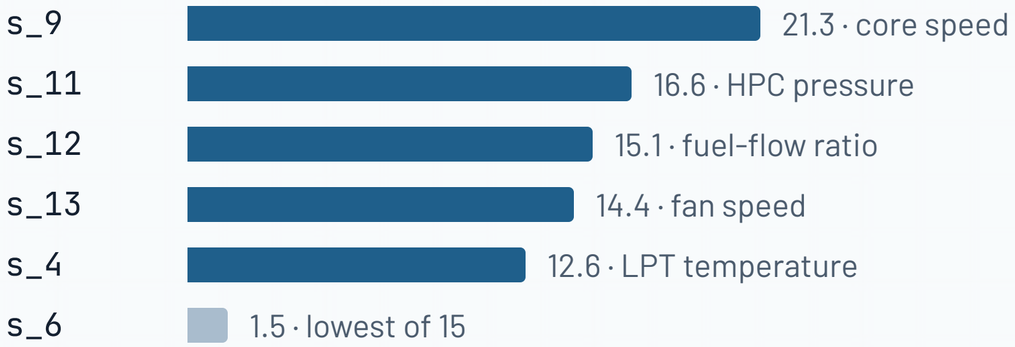
RMSE near failure for every network and engineered-feature tree

+7 to +15

RMSE cost of training any model without the cap

Near failure, the model reads the last cycles

TRANSFORMER · INTEGRATED GRADIENTS BY SENSOR (CYCLES OF RUL)



Stable across seeds (Spearman ≥ 0.86) and almost the same for the LSTM (0.97). XGBoost SHAP on the same engines: 0.54, sharing s_9, s_11, s_4. Sensor names: Saxena et al., 2008.

42%

of the attribution falls on the last 5 of 30 cycles near failure (26% early; 17% if even)

24–26%

of attention on those cycles at every stage: attention is not the explanation

≈ 0.5

Hybrid A's gate at every stage: it leans on neither branch

QUESTIONS?

On FD001, deep, hybrid and boosted-tree models tie.
The cap and the inputs matter more than the model.

Gokul VKS
Lead · trainer, LSTM, temporal CNN, Transformer, both hybrids, hybrid study

Milind Bansal
Model contract, windows, validation toolkit, FD002/FD004, validation, interpretability

Isha Tomar
Hybrid features, ablation runner and study, official test run, this deck

Yashi Agrawal
RUL cap rule, ML rerun, literature review, theory, references

Mohammad Affan Anas
PyTorch setup, CI, regularization study, dataset section, README

github.com/vks-g/cmapss-rul-hybrid